The SC-SCOPE third-cumulant endpoint closure is class-wide: MARGIN-

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

SC-SCOPE was lifted@thin-certified (2026-06-09): the third-cumulant endpoint joint is $\times 1.040 > 1$ at the anchor, sunset-saturated. The thinness raised the question whether the closure survives for ALL admissible competitors, not only the anchor / five enumerated readings. This note answers it: the joint depends only on the STEP-5B floor ρ_{lat}, and the T-017 pin $T' \leq 13$ keeps every admissible competitor well inside the closure region $T' < 60.4$. Scope: the operating endpoint $I = 2 \times 10^{-3}$, the mu^2-band $[\times 0.5, \times 2]$; the certified $R_{\max} = 0.385$ and sunset cap $S = 1.13$ are inherited.

2. MARGIN-cancellation: the joint depends only on ρ_{lat}

The certified joint (scscope-quartic-normalisation-certificate v1.0) is

$$\text{joint} = \frac{\text{MARGIN}}{C_2 + C_{\text{sunset}} + C_{\text{quartic}}}, \quad C_2 = \frac{\text{MARGIN}}{\rho_{\mathrm{lat}}}, \; C_{\text{sunset}} = \frac{\text{MARGIN}(1 - 1/\rho_{\mathrm{lat}})}{S}, \; C_{\text{quartic}} = R_{\max} C_2,$$

in which MARGIN appears in every term and CANCELS:

$$\boxed{\text{joint}(\rho_{\mathrm{lat}}) = \frac{1}{\frac{1}{\rho_{\mathrm{lat}}}(1 + R_{\max}) + (1 - \frac{1}{\rho_{\mathrm{lat}}})/S}}$$

a monotone increasing function of ρ_{lat}, saturating at $S = 1.13$ as $\rho_{\mathrm{lat}} \to \infty$ (the sunset cap). Setting joint $= 1$ gives the critical floor

$$\rho_{\text{crit}} = \frac{1 + R_{\max} - 1/S}{1 - 1/S} = 4.347.$$

Thus the endpoint closes (joint > 1) iff $\rho_{\mathrm{lat}} > 4.347$. (At $\rho_{\mathrm{lat}} = 6.55$ this reproduces the certified $\times 1.040$.)

3. The floor is $\rho_{\mathrm{lat}} = K_{\text{budget}}/(1 + T')$; critical $T' = 60.4$

The STEP-5B floor is $\rho_{\mathrm{lat}} = K_{\text{budget}}/(1 + T')$ with

$$K_{\text{budget}} = \frac{4(1 - a_0)\,\text{MARGIN}}{(\lambda' I)^2 J_{\text{eff}}} = 266.7 \quad \text{(anchor)},$$

PATTERN-INDEPENDENT (it depends on μ^2, I through the dressed propagator only; the competitor enters EXCLUSIVELY through its additive-energy richness T', the sum-circle occupancy). Hence

$$\text{joint} > 1 \iff \rho_{\mathrm{lat}} > 4.347 \iff 1 + T' < \frac{K_{\text{budget}}}{4.347} \iff \boxed{T' < 60.4}.$$

4. The T-017 pin closes the class

T-017 (res-endpoint-admissible-pin / res5-chi-link-bypass) pins $T'(Q) \leq 13$ for every admissible crystallographic-shell competitor. Since $13 < 60.4$,

$$\text{joint}(T'{=}13) = \times 1.097 > 1 \qquad (\text{margin } \times 4.6 \text{ in } T'\text{-space}),$$

and the closure is robust: (i) over the mu^2-band $[\times 0.5, \times 2]$ at the endpoint, the worst joint is $\times 1.095 > 1$; (ii) over the intensity sweep, the operating endpoint $I = 2 \times 10^{-3}$ is the THINNEST ($\times 1.097$), rising to $\times 1.129$ at $I = 4 \times 10^{-4}$. The thin $\times 1.040$ of the lift was the CONSERVATIVE $T' = n_{\text{pack}} = 40.7$ estimate; the actual admissible class ($T' \leq 13$, T-017) gives $\times 1.097$.

5. Status and honest residual

T4 STRONG EVIDENCE. The SC-SCOPE third-cumulant endpoint closure is CLASS-WIDE: joint $>$ 1 for every admissible competitor ($T' \leq 13 < 60.4$), robustly over the certified mu$^2/I$ band. This closes the last ANALYTIC H-LAYER residual identified by the T-016→T-020 arc. **Inherited / remaining:** the certified $R_{\max} = 0.385$ (the Parseval-pinned quartic, thin) and the rigorous sunset cap $S = 1.13$ are inherited from SC-SCOPE; the $T' \leq 13$ pin is T-017; the competitor-class = crystallographic-shell subsets is an operator/modeling item; anchored on $[\times 0.5, \times 2]$ (ROBUSTNESS-MU2). No tier flip: B1/B2 stay T6 on {H-LAYER} (the flip rests on the operator competitor-class sign-off plus the standing thin-certified grades).

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"MARGIN cancels in the joint ratio but enters ρ_{lat}; isn't MARGIN competitor-dependent?"* DISMISSED: MARGIN = `margin_of`(μ^2) is the ISOTROPIC LAYER margin, a function of μ^2 only – fixed across competitors. The competitor enters the joint EXCLUSIVELY through T' (its additive-energy richness), which is the content of the $\rho_{\text{lat}} = K_{\text{budget}}/(1 + T')$ parametrisation.

(β) *"$\times 1.097$ is still thin; the $R_{\max}$ certificate was hard-won."* VALID-with-inheritance: the closure inherits SC-SCOPE's thin-certified $R_{\max} = 0.385$ and the rigorous sunset $S = 1.13$; this note does NOT re-derive them. What it adds is the class-wide REDUCTION (joint $\to \rho_{\text{lat}} \to T'$) and the $\times 4.6$ T'-margin, which is much wider than the $\times 1.097$ joint suggests (the joint is a saturating function: T' would have to nearly quintuple to reach criticality).

(γ) *"Does this flip B1/B2?"* DISMISSED: no. It closes the SC-SCOPE class-wide analytic residual but leaves the operator competitor-class definition and the standing thin-certified grades. B1/B2 stay T6 on {H-LAYER}.

Quantitative sanity check (mandatory): *cancellation* – joint(ρ_{lat}) saturates at $S = 1.13$ as $\rho \to \infty$ and $= 1$ at $\rho_{\text{crit}} = 4.347$; *critical* – $T' < K_{\text{budget}}/\rho_{\text{crit}} - 1 = 60.4$; *pin* – $T' \leq 13$ gives joint $\times 1.097$, $\times 4.6$ in T'; *robustness* – mu^2-band worst $\times 1.095$, endpoint thinnest.

7. Result footer

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Result ID:                B5-BEYOND-LAYER-BOUND / SC-SCOPE
                         third-cumulant endpoint
                         class-wide closure
Precise statement:        The certified joint
                         MARGIN/(C_2+C_sunset+C_quartic) has MARGIN
                         cancel, giving
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joint(rho_lat)=1/[(1/rho)(1+R_max)+(1-1/rho)/S],
monotone in the STEP-5B floor rho_lat,
saturating at S=1.13.
joint>1 iff rho_lat>4.347. With
rho_lat=K_budget/(1+T'),
K_budget=266.7 pattern-independent, joint>1 iff
T'<60.4. T-017
pins T'<=13 for every admissible competitor, so
joint(T'=13)=
x1.097>1 (margin x4.6 in T'); worst over
mu^2-band [x0.5,x2] is
x1.095; endpoint I=2e-3 is thinnest. SC-SCOPE
3rd-cumulant
endpoint CLOSES class-wide.

Scope:	Operating endpoint I=2e-3, mu^2-band [x0.5,x2]. Inherits R_max=0.385 (certified) + sunset S=1.13 (rigorous) + T-017 pin.
Dependencies:	scscope-quartic-normalisation-certificate v1.0 (R_max, joint); scscope-floor-sharpening v1.6 (rho_lat=K_budget/(1+T')); res-endpoint-admissible-pin / res5-chi-link-bypass (T'<=13); scscope-programme-consolidation v1.0 (the lift chronicle).
Evidence grade:	ANALYTIC (MARGIN-cancellation + critical-T') + EXECUTED (res5_021_scscope_classwide_endpoint.py 5/5) + INHERITED (certified R_max, sunset cap).
Reproduction command:	python codes/vacuum/res5_021_scscope_classwide_endpoint.py
Expected output:	K_budget=266.7; critical T'=60.4; joint(T'=13)=x1.097; mu^2-band worst x1.095; 5/5 PASS.
Falsification gate:	An admissible competitor with T'>60.4 at the endpoint (would require violating the T-017 T'<=13 pin); or a re-opened R_max (>0.634) or sunset cap; or a mu^2 in [x0.5,x2] with joint<=1.
Tier before / after:	No tier flip. B1/B2 T6 on {H-LAYER}, B5 T5. SC-SCOPE 3rd-cumulant endpoint extended from the anchor/enumerated to the whole admissible class (T'<=13<60.4).
No-overclaim statement:	Closes the SC-SCOPE third-cumulant CLASS-WIDE residual (the last analytic H-LAYER piece) at the operating endpoint, but INHERITS SC-SCOPE's thin-certified R_max=0.385 + sunset S=1.13 + the T-017 pin, and leaves the operator competitor-class definition. Does NOT flip B1/B2 (those need the operator sign-off + standing grades). Anchored (ROBUSTNESS-MU2).
Next required action:	The remaining H-LAYER item is the operator competitor-class definition (admissible = crystallographic-shell

subsets up to
n_pack). With this note all analytic H-LAYER
residuals
(diagonal isotropy, chi(P) floor, off-diag 2nd
cumulant,
off-diag exchange, SC-SCOPE 3rd cumulant) are
class-wide closed.